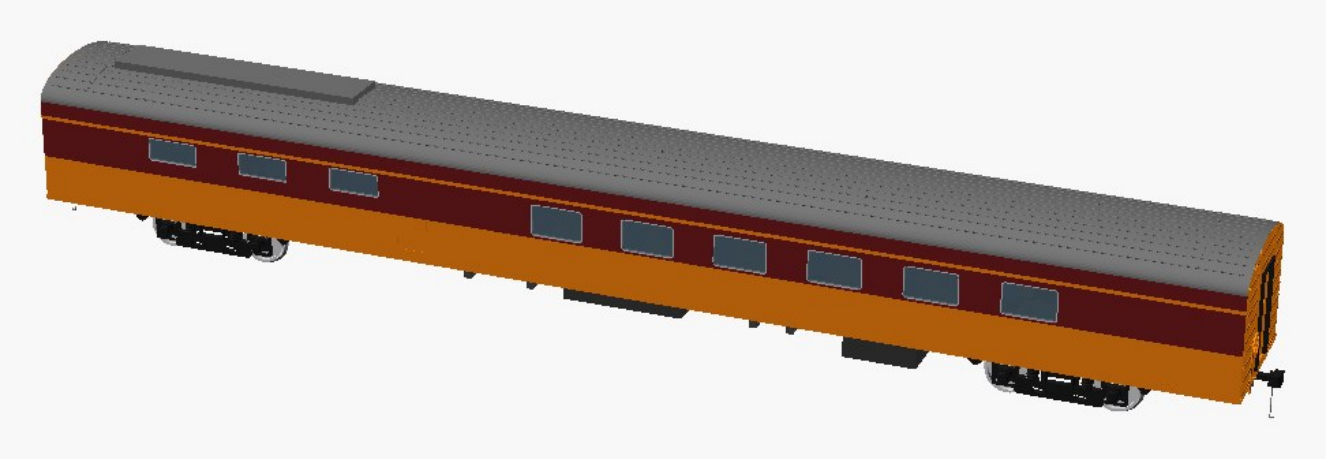
How to build the Dining Car

The Milwaukee Road | 1948 Twin Cities Hiawatha | G 1:29 · O 1:48 · HO 1:87

Created by **Tom Fleming** · n9szf@mac.com · 8 October 2026



The finished G scale model as rendered from the kit files (no lettering - that is decals or vinyl).

Diners 121-126

On May 29, 1948 - thirteen years to the day after the first Hiawatha - the Milwaukee Road put new Brooks Stevens styled Morning and Afternoon Hiawathas into service between Chicago and the Twin Cities, built almost entirely by its own Milwaukee Shops. From 1955 the fleet was repainted in Union Pacific Armour Yellow and Harbor Mist Gray, the scheme these models wear.

The Twin Cities diners, 121-126, share their body with the Olympian diners of 1947. The plan is MPL drawing A-4836: kitchen at the A end with high oblong windows and short service doors, six 3'-8" dining windows a side, a steward's compartment at the B end and a kitchen exhaust housing on the roof.

To check: window and side-door heights (estimates); number 121 is the default.

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What is in the kit

Every scale has the same layout inside the *Kit* folder. Nothing has lettering on it, inside or out, except the small part number engraved on the hidden inside face of each section (for example *B2/4* = body section 2 of 4, arrow to the A end).

Folder	What it holds
scale / OpenSCAD	One stand-alone OpenSCAD file for the car at that scale - everything in a single file.
MR Print Files (top of Fleet export)	The program that makes the print files for this car on your own computer: choose train set, car, scale, printer and colour. It needs OpenSCAD (free). Double-click *MR Print Files.command* on a Mac, or run *MR Print Files.py* with Python.
What it makes: one colour (STL)	Every part in one colour, cut into pinned sections that fit a 256 mm bed (or body, roof and floor in one piece each for an 800 mm bed).
What it makes: colour (3MF)	Body sections as ONE object with the colour parts already on separate filaments; window inserts with rim and glass; the other parts as single 3MF files.
scale / Pictures	Snapshots of the rendered model used in this book.

Scale	Car length	Bambu P1S	OrangeStorm Giga	Kit contents
G 1:29	871 mm	4 body + 4 roof + 5 floor section(s)	body, roof, floor in one piece each	full kit: body, roof, floor, interior, step boxes, window inserts, lighting-pack box
O 1:48	526 mm	3 body + 3 roof + 4 floor section(s) (laid across the bed diagonal)	body, roof, floor in one piece each	shell kit: body, roof, floor, window inserts
HO 1:87	290 mm	1 body + 1 roof + 1 floor section(s) (laid across the bed diagonal)	not needed - prints whole on the P1S	shell kit: body, roof, floor, window inserts

Trucks are a separate project and are not in this kit. HO and O are first-pass shell kits that have not been test-printed yet: print one body section and check the fits before committing to a whole car.

Hardware

Scale	You need
G 1:29	M2.5 x 10 self-tapping screws (floor); 6 x 3 mm disc magnets (roof); 1.75 mm filament pins; Kadee #1906 couplers; LED strip + lighting pack (optional)
O 1:48	M2 x 6 self-tapping screws (floor); 5 x 2 mm disc magnets (roof); 1.75 mm filament pins; O scale passenger trucks (your choice) on 4-40 / M2.5 screws; Kadee O couplers (#805 type box)
HO 1:87	M2 x 5 self-tapping screws (floor); 3 x 1 mm disc magnets (roof); HO passenger trucks (your choice) on 2-56 screws; Kadee #5 / #148 couplers in their own boxes

Printing on the Bambu P1S with the AMS

- **Filament.** PETG. Load the AMS as: **1** Milwaukee Road orange, **2** maroon, **4** clear or smoke for the window glass (**3** is not used).
- **Settings.** 0.2 mm layers, 3 walls, 240-250 C nozzle, 70-80 C textured PEI, no supports. Body sections stand upright exactly as they open; roofs stand on their lip; floors print upside down.
- **Colour body sections.** Make the files with *MR Print Files* (Print in colour) and open a body file. Bambu Studio says 'The 3mf file has invalid config, load geometry data only' - click OK, that is expected. The section arrives as one object with two parts already on filaments 1 and 2. Do not move the parts apart.
- **Why it prints cleanly.** The sides stand upright, so the orange lower side (to 33.6 mm), the maroon window band (to 62.6 mm), the orange stripe (to 65.2 mm) and the maroon letterboard are whole layers - only four filament changes. The car ends are orange like the sides, so the end sections change colour only where the maroon bands wrap round the corners.
- **Roof, floor, interior.** One colour each (roof gray, floor and underbody dark gray or black).

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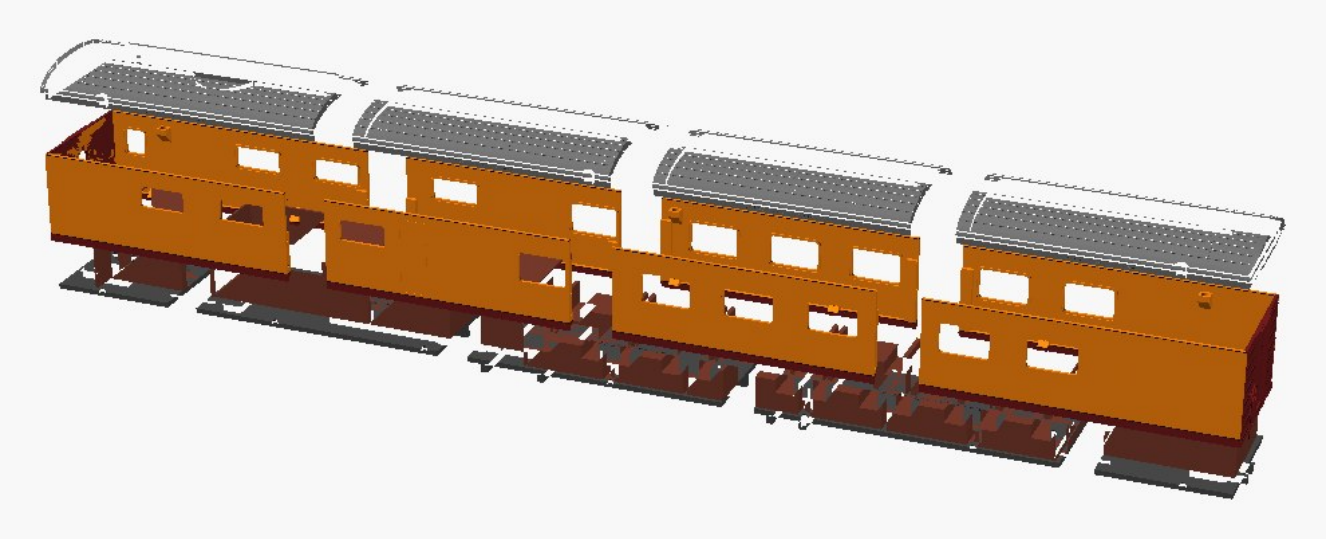
- **Window inserts.** *windows_colour.3mf* has the rims on filament 2 and the glass on filament 4. The inserts come as one grid; if it is bigger than the plate, right-click, Split > To objects, then Arrange. Print two or three first and try them in a printed side.
- **One colour instead.** Print everything from the plain files and paint: prime, Milwaukee Road orange sides and ends, maroon window band and letterboard, gray roof.

Printing on the Elegoo OrangeStorm Giga

- The *OrangeStorm Giga* folder has the body, roof and floor each in a single piece - no joints, no pins. The G scale body is 871 mm long and lies across the bed diagonal as it opens.
- The Giga prints one filament at a time, so there are no multi-colour files for it. Because the colours are whole layers you can still get the bands with filament changes at these heights (add a pause / M600 in the slicer): **G scale** orange from the bed to 33.6 mm, maroon to 62.6 mm, orange to 65.2 mm, maroon to the top. **O scale** orange to 20.4, maroon to 37.8, orange to 39.4 mm, maroon to the top. The car ends come out banded this way; Milwaukee ends were plain orange - paint the maroon off them, or leave it.
- A body this long needs a clean, level bed and a brim at the ends; PETG at 0.2-0.28 mm layers with a 0.6 mm nozzle is a good starting point.

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Building the car

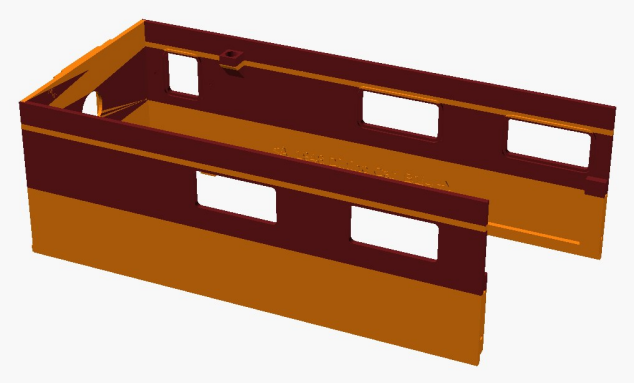


Every printed section pulled apart: roof sections above, body sections in their print colours, floor and interior below.

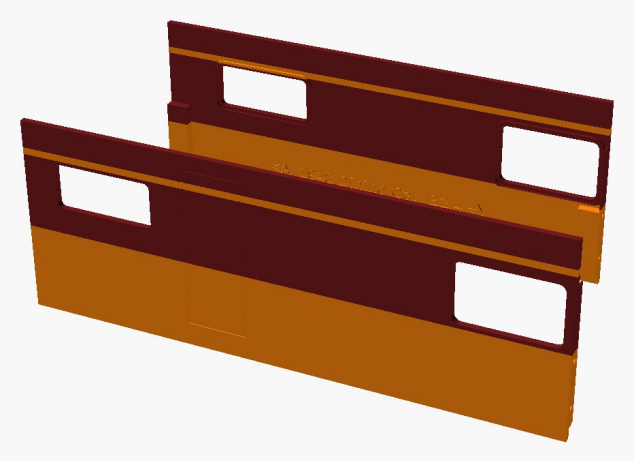
1 **Sort the parts.** Lay the sections out by their engraved numbers, A end to the left: body B1, B2 ..., roof R1 ..., floor F1 .... The two end body sections are U-shaped; the middle ones are separate -Y and +Y side walls.

2 **Clean the pin holes.** Run a 2.1 mm drill through every pin hole by hand and cut pins from 1.75 mm filament, 14 mm long. Dry-fit everything before any glue.

3 **Join the body.** On a flat surface pin and glue one side wall at a time, starting from B1 at the A end (CA with activator or 2-part epoxy - solvent cement does not weld PETG). Keep the outside faces flush across each joint so the paint bands line up.



An end section (B1): U-shaped, with the orange car end.



A middle section: the two side walls print side by side.

4 **Join the floor.** Pin and glue the floor sections (two pins per joint). The floor joints are staggered from the body joints, so the floor splices the body when it goes in.

5 **Underbody equipment.** The battery boxes, tanks and other equipment are separate parts (*equipment* file, printed top face down). Each has two holes that match holes in the frame: push a short 1.75 mm filament pin into each and glue the part on. (Set EQUIP_FIX to *screw* in the OpenSCAD file for M2.5 screws down through the floor instead, or untick EQUIP_SEPARATE to print the equipment on the frame.)

6 **Fit the floor.** Drop the body over the floor so the ledge inside the walls sits on it, and screw up through the floor into the wall bosses. Do this before the glue in the side walls has fully cured - it squares the car.

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7 **Windows.** Press the inserts in from inside, flange first. Too tight: raise WIN_FIT in the OpenSCAD file; too loose: a drop of clear glue.

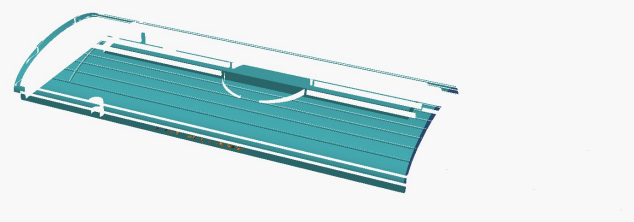
8 **Roof.** Pin and glue the roof sections. Glue the magnets into the roof bosses and the blocks at the top of the body - check the polarity so the roof pulls down. The roof stays removable.

9 **G scale extras.** Glue the interior sections to the floor, screw the step boxes up under the vestibule doors, fit the lighting-pack box and LED strip (see the Lighting Pack notes), bend brass-wire grab irons for the holes beside the doors.

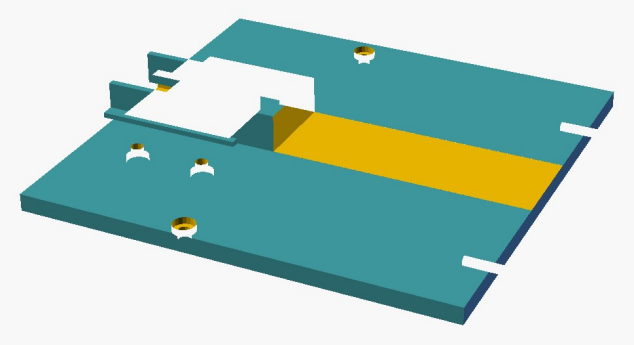
10 **Trucks.** G scale trucks have their own kit in *Fleet export / Trucks* (2 sideframes, 1 bolster, 2 end beams and printed wheels per truck, F683ZZ bearings, 3 mm brass axles; see the truck's own how-to-build); O and HO use bought trucks. The bolster pads under the floor take the pivot screw.

11 **Couplers.** Each end of the floor has a printed pocket - a cheek either side and a stop behind. Drop the Kadee gearbox in from below, fix it with its screw and set the height with a coupler gauge.

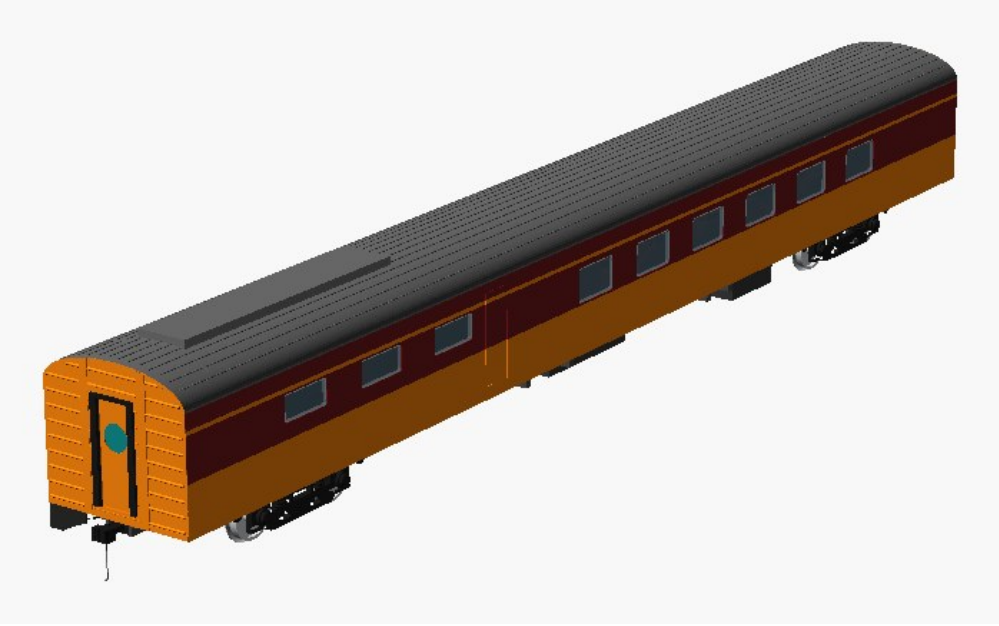
12 **Lettering.** Nothing is printed on the car. Add THE MILWAUKEE ROAD on the letterboard and the car number with decals or cut vinyl once the paint or filament colour is finished.



A roof section, standing on its lip as it prints.

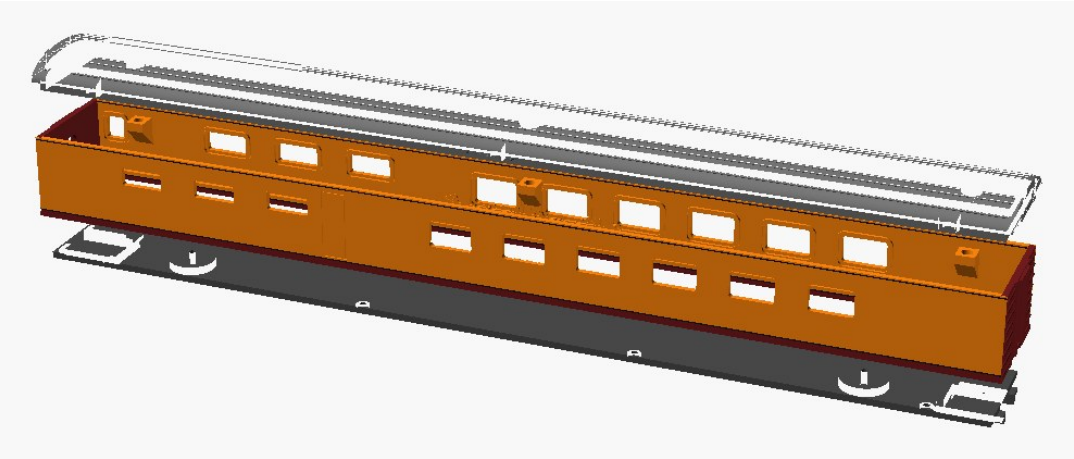


A floor section, upside down as it prints.



The A end: end sheet in the body colour, end door with its porthole, diaphragm frame.

HO and O scale notes



The HO shell kit: body, roof and floor each print in one piece across the P1S bed diagonal.

- The smaller scales are **shell kits**: body, removable roof, floor and window inserts. The walls, roof lip, magnets, screws and window fits are sized for each scale rather than shrunk from G.
- **HO** (290 mm): prints whole on the P1S, laid diagonally. **O** (526 mm): two or three pinned sections on the P1S, or one piece on the Giga.
- **Trucks and couplers are yours to choose.** The bolster pad height, pivot screw hole and coupler pocket are first guesses for common commercial parts. Measure yours and change BOLSTER_H, TRUCK_SCREW_D and COUPLER_BOX near the top of the stand-alone OpenSCAD file, then export the floor again.
- These two scales have not been test-printed yet. Print one body section and a few window inserts first.

Looking at the model

Open the stand-alone file and use the Customizer. **Show in the assembly view** has a tick box for the body, roof, floor, interior, windows, steps, trucks, couplers, light box and underbody equipment. **Paint** switches between MILW (orange and maroon - the default), UP (Armour Yellow and Harbor Mist Gray) and none (plain, as printed in one colour). **Car frame** has FRAME_ONE_PIECE (the floor as one part - needs the Giga), EQUIP_SEPARATE and EQUIP_FIX (pin or screw).

Changing the model

Each *OpenSCAD* folder holds one stand-alone file - open it in OpenSCAD (2025 or newer), pick PART, SECTION, PRINTER and COLOR_PART in the Customizer, press F6 and export (File > Export > STL). Everything the file needs is inside it - no other files or folders are required.

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